Methodology report: What Normality Can and Cannot Do in Theories of Actual Causation

Anonymized production account for the manuscript revised 28 October 2026.

Models, roles and production history

The manuscript began as a Claude-authored draft in an existing AI research project and was subsequently revised by an OpenAI Codex assistant. The project record names a Claude Opus 5.5 orchestrator operating through Claude Code 2.1.306. This is a report of the recorded environment, not an independently verified model identifier for every historical drafting call. An exact served model version for each Codex session is not established by the available record.

Claude's workflow generated candidate topics, developed arguments, wrote drafts and produced synthetic referee reports. Codex conducted literature comparisons, mathematical checks, adverse self-review and substantive revision. The final revision on 28 October 2026 used one Codex agent. It launched no additional agents, no paid model-review pipeline and no new independent Claude, Gemini or Opus review. Simulated philosophical perspectives are evaluation techniques, not endorsements by the philosophers concerned.

This manuscript is the third version in its recent review lineage. The initial Claude draft, the 25 October review edition and the present source are preserved separately.

Topic generation and selection

The project explored a register of 120 idea cards, filtered through groups of 40, 16 and eight, as well as a separate set of six ambitious problems. A later inventory distinguished 13 developed tracks covering approximately ten arguments and eight exploratory drafts. These are artifact and topic counts, not counts of independent discoveries. They summarize the recorded search; a complete transcript-based count of every attempted idea is unavailable.

Selection favored a definite claim about published work, a proposed increment over close literature, and an argument that could survive explicit objections. The final pass prioritized the accuracy and free-evidence essays, prepared the causation essay as a conditional third candidate, and retained the withdrawal of a separate flattening essay after adverse review. Selecting a candidate does not certify its originality or philosophical importance.

Instructions and human involvement

The human requested AI-originated philosophical essays, independent originality assessment, more significant arguments, multiple candidates, rigorous criticism, revisions incorporating other AI reviews, and preservation on GitHub. The human arranged access to sources and carried some material between systems. The human stated that substantive philosophical feedback was being withheld for the contest. No new human philosophical argument was solicited or used in this final revision.

Representative instructions in the visible project conversation were to check originality independently, generate and gradually filter at least 100 more ideas, hold the work to high standards beyond the judging panel, and review and revise as autonomously as possible before the deadline. These were workflow and quality requests. The models selected the particular proofs, constructions, objections and revisions described below.

The research scaffold required one precise thesis, source-faithful criticism, explicit bridge assumptions, strong objections, verified references, and separation of proof, finite computation and unresolved conjecture. No hidden instructions to reviewers or evaluation systems were included in the essays. The final text, including appendices and footnotes, was counted against the word limit; bibliography entries alone were excluded.

Retrieval, context and provenance

Available tools included GitHub retrieval and versioned publication, web retrieval, local files and mathematical computation, document typesetting, and persistent artifact storage. The authoring conversation also contained earlier conversational context. Personal-context search was not used to generate ideas in the final pass. Availability of prior or personal context is disclosed; this was not an independently certified isolated run. Unrelated private material is not reproduced here.

An earlier project note raised a context-exposure concern. A subsequent limited audit of project instructions and recorded human inputs identified no human-supplied philosophical contribution in those inspected records. That correction did not establish the contents of every historical session or erase the earlier disclosure. The supported distinction is between exposure to context, an identified substantive contribution, and an essay's philosophical quality. None is automatically evidence of the others. This report preserves that limitation without claiming either a complete clean-room audit or a finding of disqualification.

Full originating prompts and transcripts are not all present in the inspected repository. The custodian should retain the chats and existing project records. This report provides the available production account, not a reconstructed transcript. Account identifiers, repository-owner information and unrelated private exchanges have been omitted from this public-facing report.

Evaluation and resources

The final pass reconstructed the arguments, checked exact assumptions and counterexamples, compared relevant primary literature, edited overstatements, and used targeted computations to probe new claims. Computation supplements the written proofs; finite checks are not literature searches or independent expert endorsement. Earlier checks not rerun in this pass are distinguished from new checks in the manuscript-specific account.

The workflow used subscription-based assistants and local tools. Exact cumulative machine time, human time, tokens and financial expenditure across the project could not be established. Requested budget limits and subscription resets are not treated as measured usage. No contest submission was performed by the final revision agent. No human adjudication of eligibility or novelty is claimed.

Work on this manuscript

The source is the Claude draft on normality filters in actual causation, followed by the 25 October revision. It is not the distinct exploratory manuscript about endless causal refinement. The earlier revision repaired the frozen-witness lemma using acyclic uniqueness and AC3 minimality.

The present pass rechecked the core claims against Halpern's 2015 paper and Halpern and Hitchcock's normality paper, using the identified preprint versions for pagination. It corrected the presentation of their framework: the componentwise conformity filter analyzed

here is a strict implementation, while their actual account explicitly permits exceptions and richer rankings. It clarifies that root variables remain intervenable, and that the conformity test concerns non-root equations. A reversed tuple in the short-circuit proof was corrected.

The final computation expands the central case comparison from 75 weak orders to all 355 preorders of the four categories. It examines every nonempty candidate subset, including candidates containing the outcome, and applies minimality after normality filtering. No mismatch was found in 2,130 calibration comparisons or 3,195 paired-case comparisons. In these cases the stipulated inertial coding coincides with event coding. The independent proof remains conditional on the stated class, codings and causal judgments.

The philosophical revision makes the isomorphism result an explanatory constraint. Distinguishing identical represented structures requires additional information or a different representation. It no longer claims that every richer normality theory fails, that all principled default codings were tested, or that coarse-grained causal judgments are indisputable.

Remaining limits: Halpern's 2016 Actual Causality, especially chapters 3–4 and the requested earlier pages, was not reliably accessed in full. The manuscript does not attribute a position to that unread book. Comparison against it is still needed before a strong originality judgment. Beckers's normal-backup challenge and the general isomorphism problem are acknowledged prior work. This candidate is lower-confidence than the other two on novelty and interpretation.

Final files and length

The matching essay is titled What Normality Can and Cannot Do in Theories of Actual Causation. Its source count is 5,793 whitespace-separated tokens after markup removal, counting title, abstract, headings, tables, mathematical tokens, appendices and footnotes while excluding the bibliography. This conservative count is below 6,000. The matching PDF is the version produced from that source. This report has no word-limit claim.